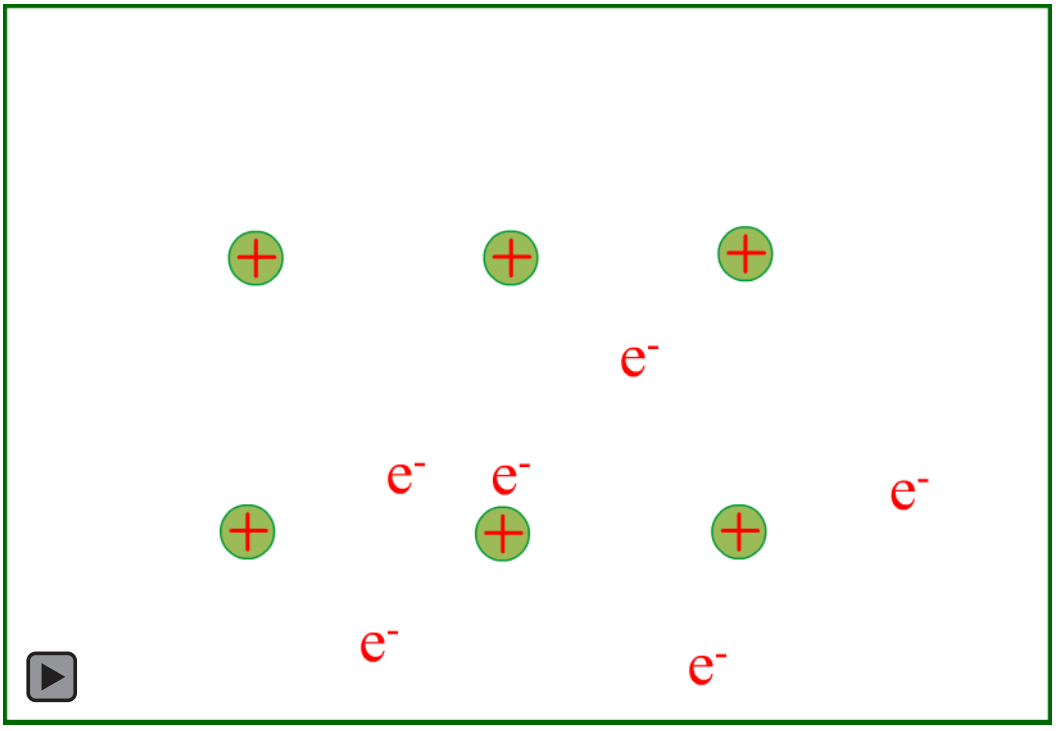
Class 5: Free Electron Gas?

Electronic conductivity, as discussed earlier, is of scientific as well as technological interest. It is of scientific interest since it varies over 24 orders of magnitude across various materials, and it is of technological interest since many commonly used gadgets depend on electronic conductivity as an essential step for their functioning. In the present as well as upcoming classes several models will be built to examine how the constituents of the materials interact with each other and how that results in the conductivity displayed by the material. In particular we will focus on metals which display a positive thermal coefficient of resistivity.

It is worth noting that in the immediate discussions we will not include superconductors. Superconductors exhibit phenomena that are quite unique to them and are not seen in typical metallic conductors at room temperature. Superconductors are covered in greater detail in a later class.

Metals behave as though there is a 'free electron gas' inside them. In this class we will examine the idea of a 'Free electron gas' in considerable detail and specifically look at how reasonable or unreasonable such a 'model' for a metal is.

The 'free electron gas' picture is as follows: Atoms in metals occupy regular lattice sites. These atoms have a natural valence state and consistent with this valency, they release electrons and become positively charged ions. The lattice sites are therefore occupied by ionic cores. The released electrons are not confined to any single ionic core, but are free to roam the extent of the solid. These electrons are therefore referred to as 'free electrons' or as the 'free electron cloud'. The free electrons are however confined to within the extent of the solid – they have not escaped the physical boundary of the solid. In this sense they are said to be ***localized*** within the extent of the solid. A schematic of metallic solid containing the free electron gas is shown in Figure 5.1 below.



Animation of figure 5.1

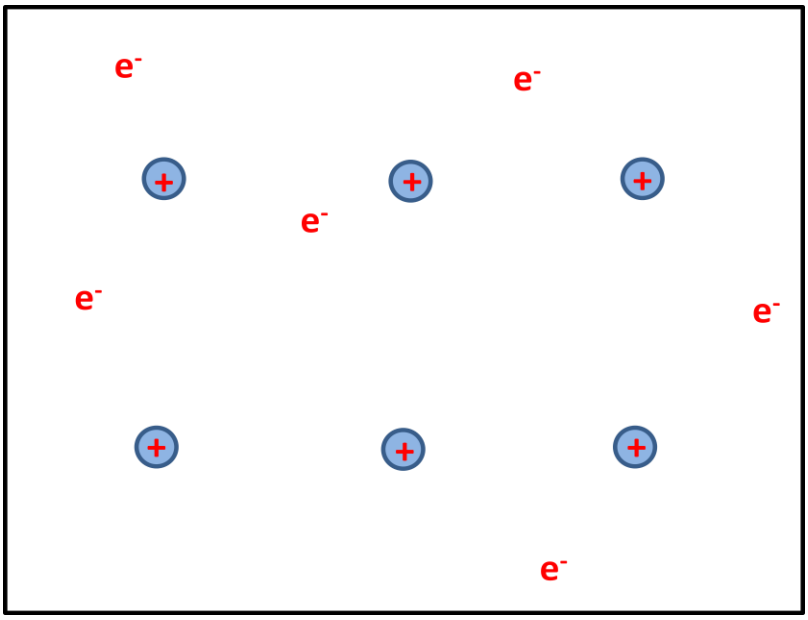


Figure 5.1: A schematic showing a metallic solid containing ionic cores at lattice sites and a free electron gas roaming through the solid

For example, if there are 1000 atoms in a sample of a metal, and if the natural valency of the atoms is +1, then each of the atoms releases one electron to form the free electron gas, which therefore consists of 1000 electrons. The 1000 free electrons are free to roam through the extent of the metallic sample. The remaining electrons per atom, remain in the vicinity of each individual ion and are not free to roam through the extent of the solid. The positively charged ionic cores will repel each other, and left to themselves, the solid will fall apart. The negatively charged free electron cloud provides the negatively charged atmosphere within which the positively charged ions can sit stably.

Since the electrons are free to run within the extent of the solid, they can be thought of as atoms of a gas confined within a box the size of the solid.

Given the above picture of the solid, or the above model for the solid, we will now put together equations and numbers relevant to the above picture, and see if from these equations and numbers we are able to extract predictions about the conductivity of the solid. To do this let us first focus on the phrase ‘free electron gas’. In this phrase, the word that is important for our immediate discussion is ‘gas’. When we discuss gases, we usually begin with the familiar ‘Ideal gas’. We are familiar with the equations associated with the ideal gas and the behavior of the ideal gas. In the discussion in the next few classes, we will impose the ideas, rules, and concepts associated with an ‘ideal gas’ on the ‘free electron gas’. The thought that concepts associated with a gas can be extended to a solid, does not seem very well founded at first glance. In fact, as a state of matter, we treat a solid as being distinctly different from a gas. Therefore, in this class we will examine the validity of the idea that ideal gas rules can be extended to one of the constituents of a metallic solid, namely the free electrons. Even at this stage we must note that we are not attempting to treat the entire solid as being equivalent to the gas. Only the

free electrons in the solid, which are running freely through the extent of the solid, are being compared with the atoms of a gas, which run freely through the extent of the container that holds them. There is, therefore some similarity between the circumstances faced by the free electrons and ideal gas molecules. We are taking advantage of this similarity to extrapolate ideas from ideal gases to free electrons.

What aspects of this extrapolation are easily justifiable? What aspects of this extrapolation are causes for concern? We will now attempt to answer these questions.
Assume the solid has a simple cubic crystal structure. There are atoms at the corners of the simple cube as shown in Figure 5.2. The atoms have a radius 'r', and the crystal structure has a lattice parameter 'a'.

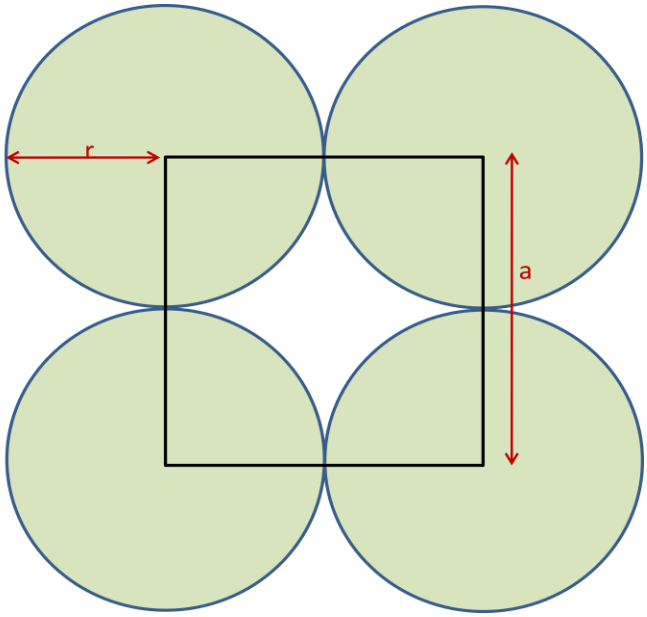


Figure 5.2: Side view of a simple cubic crystal

The volume of the cube is: a^3

There are 8 corner atoms, each shared by 8 adjacent unit cells, therefore these contribute 1 atom to the unit cell on average. Therefore the volume of the atom associated with the cube is: $\frac{4}{3}\pi r^3$

From Figure 5.2 it can be seen that $r = \frac{a}{2}$

The packing fraction is therefore: $\frac{\frac{4}{3}\pi\frac{a^3}{8}}{a^3}$

Simplifying, this fraction evaluates to approximately 50%. In other words a metal having a simple cubic crystal structure is 50% empty!

In a more closely packed structure such as a Face Centered Cubic (FCC) structure, as shown in Figure 5.3,

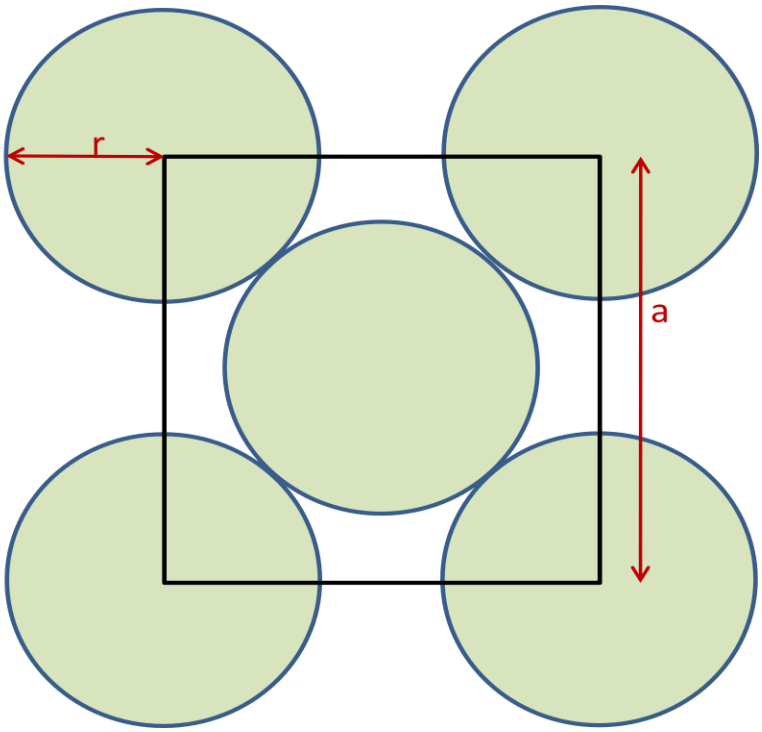


Figure 5.3: Side view of a face centered cubic crystal

There are 8 corner atoms, each shared by 8 adjacent unit cells, therefore these contribute 1 atom to the unit cell on average. Additionally, there are 6 face centered atoms, each shared by two adjacent unit cells. These contribute a total of 3 atoms to the unit cell on average. Therefore there are 4 atoms per unit cell.

Here $4r = a\sqrt{2}$

For FCC, there are 4 atoms per unit cell.

Therefore, packing fraction is given by $\dfrac{4 \times \frac{4}{3}\pi\left(\frac{a\sqrt{2}}{4}\right)^3}{a^3}$; which is approximately 71%.

BCC has a packing fraction of approximately 68%

Therefore metallic objects are approximately 30% to 50% vacant. From the perspective of this available empty space within the solid, and taking into account the extremely tiny

size of electrons relative to atoms, it is not unreasonable to compare the state of free electrons in a solid, with that of the atoms of an ideal gas in a relatively large vessel.

What are reasons for caution with this model?

Consider the number density of particles in a solid and that in an ideal gas

At STP (0 ^{o}C and 1 atmosphere pressure), 1 mole of an ideal gas occupies 22.4 l

1 mole = 6.023 X 10^{23} atoms

Therefore 6.023 X 10^{23} atoms are present in 22.4 l at STP or in 22.4 X 10^{-3} m^3

On a 1 m^3 basis, a gas therefore has 6.023 X 10^{23} / 22.4 X 10^{-3} atoms, which is approximately 3 X 10^{25} atoms/m^3.

Consider the metal silver, Ag, which has a density, ρ, of 10.5 gm/cm^3, atomic mass of 107 amu, and usually demonstrates a valency of +1.

ρ = 10.5 X 10^6 gm/m^3

Therefore the number of moles of Ag per m^3, is given by:
10.5 X 10^6 / 107 moles/m^3

And the number of atoms/m^3 is given by
6.023 X 10^{23} X 10.5 X 10^6 / 107 = 6 X 10^{28} atoms/m^3

Since we will assume that the atoms will be univalent, this is also the number of free electrons per m^3.

In summary, from the above calculations, we find that there are approximately 10^{25} particles per m^3 in an ideal, whereas there are 10^{28} particles per m^3 in the free electron gas. Which means there is a 3 orders of magnitude increase in the particles per m^3 in a free electron gas with respect to that in an ideal gas. In other words, the free electrons are a thousand times more densely packed than atoms of an ideal gas. This is the reason we need to be cautious when we extend ideal gas behavior to the free electrons in a solid. In an ideal gas we assume that the particles do not interact with each other between collisions. The more densely the particles are packed, the less reasonable it is to state that the particles do not interact between collisions.

Therefore, the vacant space within the solid makes the ideal gas – free electron gas comparison reasonable, while the significantly higher particle density in the free electron gas gives us reason for caution in this comparison.

With these ideas in mind let us now list some rules that we will expect the free electron gas to obey, which are similar to the rules that are expected to be obeyed by ideal gas atoms, and develop what is called the 'Free electron theory for metals'.

1) Electrons undergo collisions with each other, which are instantaneous and these lead to scattering.

As electrons run through the material, they collide with other electrons. We are assuming that the collisions occur without any significant time being associated with the act of the collision

2) Between collisions, interactions with other electrons, and ionic cores is neglected in its details. However we do indicate an averaged resistive term to account for the interaction with the rest of the material.

In reality, each of the electrons will experience the negative charge of the rest of the electrons in the free electron cloud. However, the negative charge of the electron cloud is balanced by the positive charge of the ionic cores, and therefore it is not entirely unreasonable to neglect interactions with the rest of the material in between collisions.

3) The mean free time between collisions, τ, for the free electrons, is independent of position and velocity of the electrons.

The electrons collide with other electrons as they move around. If an electron is moving fast, there is a greater chance that it will collide with another electron sooner, when compared to an electron moving slowly. However, it is impossible to keep track of every individual electron between every two collisions. Therefore an averaged time between collisions is assumed and used for the entire collection of electrons. Since it is averaged, we use it without regard to the position and velocity of any specific electron

4) Electrons attain equilibrium with their surroundings through collisions with other electrons

When two electrons collide, there is an exchange of energy. It is through this process of exchange of energy that the collection of electrons, over a period of time, attains equilibrium with their surroundings. For example, when a block of metal that is cold, is placed in a room that is warm, part of the process by which the temperature of the block reaches that of the room is that the free electrons close to the surface of the block experience the higher temperature and gain velocity and subsequently collide with free electrons in the interior of the solid and pass on the energy to those electrons as well. This process continues till all the free electrons in the solid are moving with velocities that are consistent with the increased temperature of the solid. In this context we must also note that there is always a distribution of velocities amongst the electrons and not a single velocity for all of the electrons. Taken in its entirety, the distribution will be consistent with the temperature of the solid. For ease of calculations we can work with the mean of

these velocities. (The ions also participate in the transfer of heat into the solid, but it is not a process that we will focus on in the immediate discussions).

With the above rules, we will examine electronic conductivity and thermal conductivity of metallic solids. In this regard, the mathematical manner in which we employ the rules that we have stated above, is based on the mathematical treatment we employ for ideal gases. Therefore, in the next class we will briefly examine the mathematical treatment of ideal gases, and highlight the results we will take from that analysis for use with the free electron gas. In the classes that follow, we will use these results to predict electronic conductivity and thermal conductivity of metals as dictated by the above 'Free electron theory' of metals.